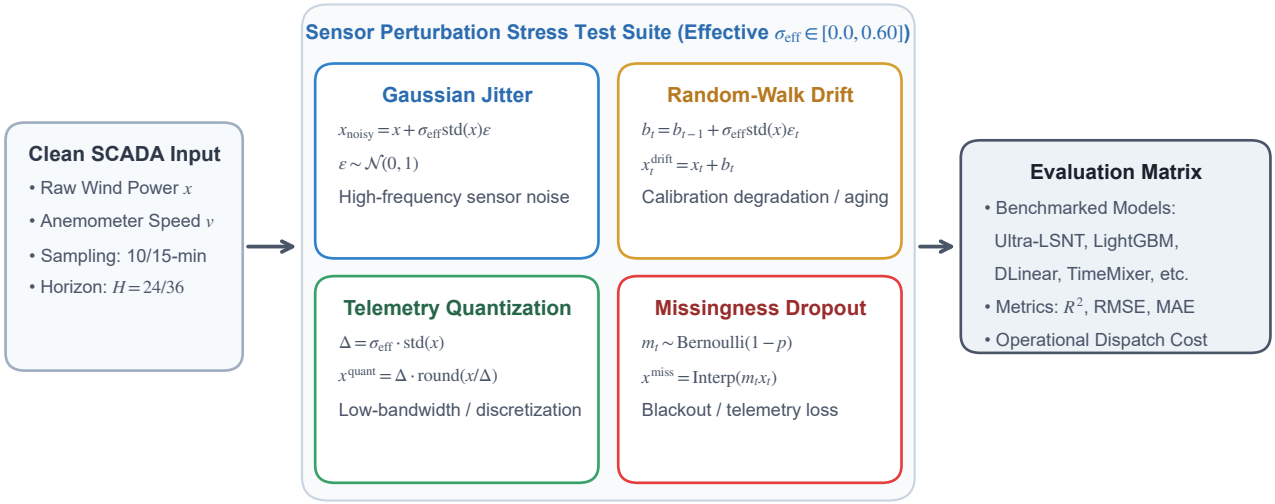

Clean SCADA Input
• Raw Wind Power x
• Anemometer Speed v
• Sampling: 10/15-min
• Horizon: $H = 24/36$
Sensor Perturbation Stress Test Suite (Effective $\sigma_{\text{eff}} \in [0.0, 0.60]$)
Gaussian Jitter
$x_{\text{noisy}} = x + \sigma_{\text{eff}} \text{std}(x) \varepsilon$
$\varepsilon \sim \mathcal{N}(0, 1)$
High-frequency sensor noise
Random-Walk Drift
$b_t = b_{t-1} + \sigma_{\text{eff}} \text{std}(x) \varepsilon_t$
$x_t^{\text{drift}} = x_t + b_t$
Calibration degradation / aging
Telemetry Quantization
$\Delta = \sigma_{\text{eff}} \cdot \text{std}(x)$
$x^{\text{quant}} = \Delta \cdot \text{round}(x/\Delta)$
Low-bandwidth / discretization
Missingness Dropout
$m_t \sim \text{Bernoulli}(1-p)$
$x^{\text{miss}} = \text{Interp}(m_t x_t)$
Blackout / telemetry loss
Evaluation Matrix
• Benchmarked Models:
Ultra-LSNT, LightGBM,
DLinear, TimeMixer, etc.
• Metrics: R^2, RMSE, MAE
• Operational Dispatch Cost